FIRE *the weekly fire alarm test is 10am on Wednesdays*

- Follow green ‘running man’ signs to the fire escape routes signposted ‘Fire Exit’ with an arrow pointing to the nearest exit
- Close doors behind you
- Let us know if you need assistance to leave the building
- Assemble on the grassy area at the front of the Nucleus until told it is safe to re-enter the building





DON’T



Please confirm your attendance here (for Graduate school):



THE UNIVERSITY of EDINBURGH

Welcome to

Dealing With Data

SBS PhD Induction – October 2026

Livia Scorza – BioRDM Data team lead
Daniel Thédié – BioRDM Research software specialist
Juliana Rodríguez Cubillos – BioRDM PhD Candidate: AI / metadata

CC BY

bio_rdm@ed.ac.uk



wooclap

Event code

ZPIIYMA

Why are we here?

Your data, your responsibility

Learn how to manage data so you and others can:

- Find your data easily when needed
- Understand what you did in the past and reproduce if needed
- Avoid data loss and duplicated data
- Have datasets that are ready to publish



Data must be available to all

- Publish data that is reproducible and reusable

Course overview



- Open Science and FAIR principles
- Collecting FAIR data
- Managing Interoperable data
- Managing FAIR data records
- Planning ahead: writing a DMP
- Data-related resources at the University



Open Science and FAIR principles

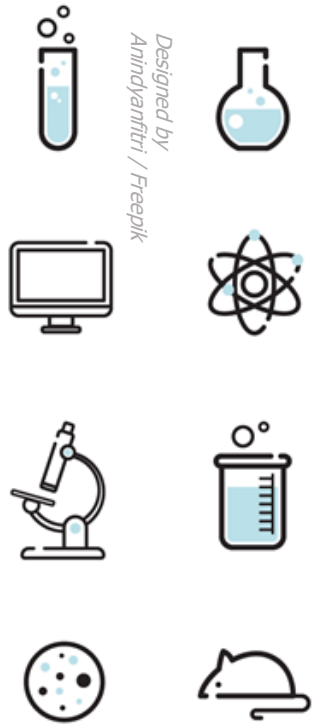
Sharing research data as a goal

What is data?

Data can be defined as facts, figures, or information that is collected, used and stored for reference or analysis

Biological data is diverse!

- **Information about physical samples**:
  - animal/ plant (live, cell cultures, frozen tissues, etc)
  - patient information
  - bacterial / yeast stocks
- **Quantitative / numerical data**: Spreadsheets (tables), csv files
- **Images** (pictures, microscopy images, 3D scans)
- **Code** (Software programs, analysis codes, algorithms)
- **DNA, RNA, protein sequences** (plasmids, NGS sequencing files, etc)
- **Protocols** (recipes, laboratory and measurement protocols)



Open Science

Open science is the movement to make scientific research outputs **accessible to all levels** of society

- Publications
- Datasets
- Software / code
- Metadata



- general public
- amateurs
- professionals
- students
- teachers

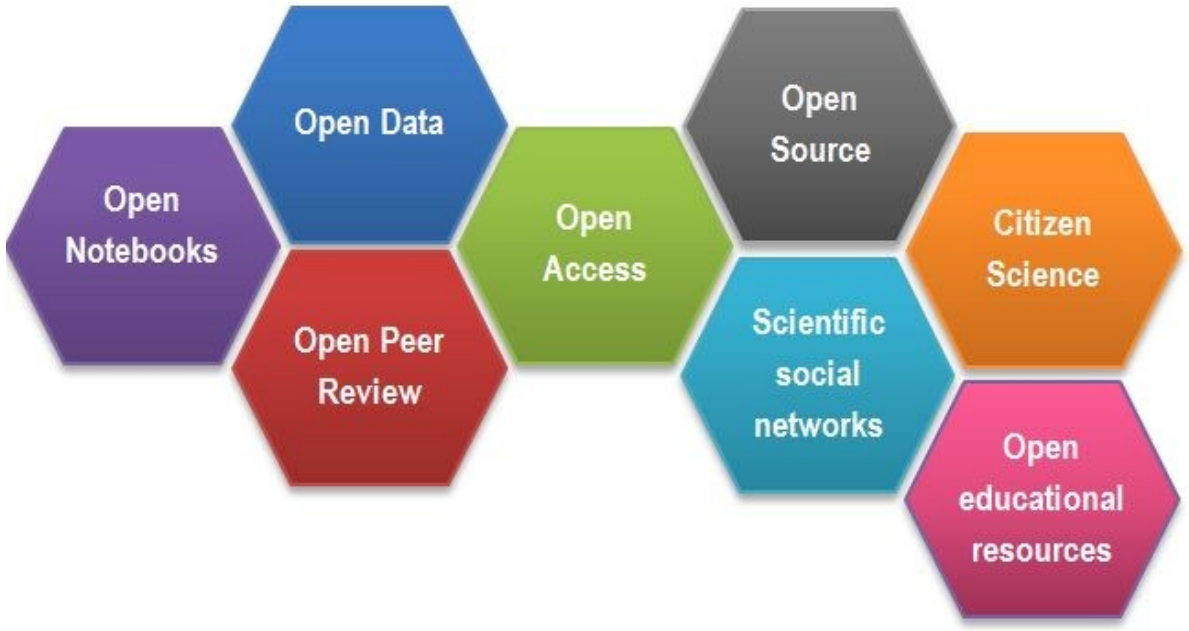


Digital technologies for diffusing knowledge

Open Science

As Open as Possible, as Closed as Necessary

Publicly funded data should be shared with minimal restrictions



CREDITS *Open Science facets as a beehive* by Gema Bueno de la Fuente licenced under CC-BY

Exceptions

- Sensitive data
- IP



Common Issues in Data Sharing



No numerical data available / only averages



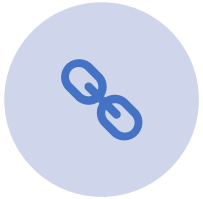
Incomplete datasets



No / little supporting information about the data



Vendor-specific file formats.

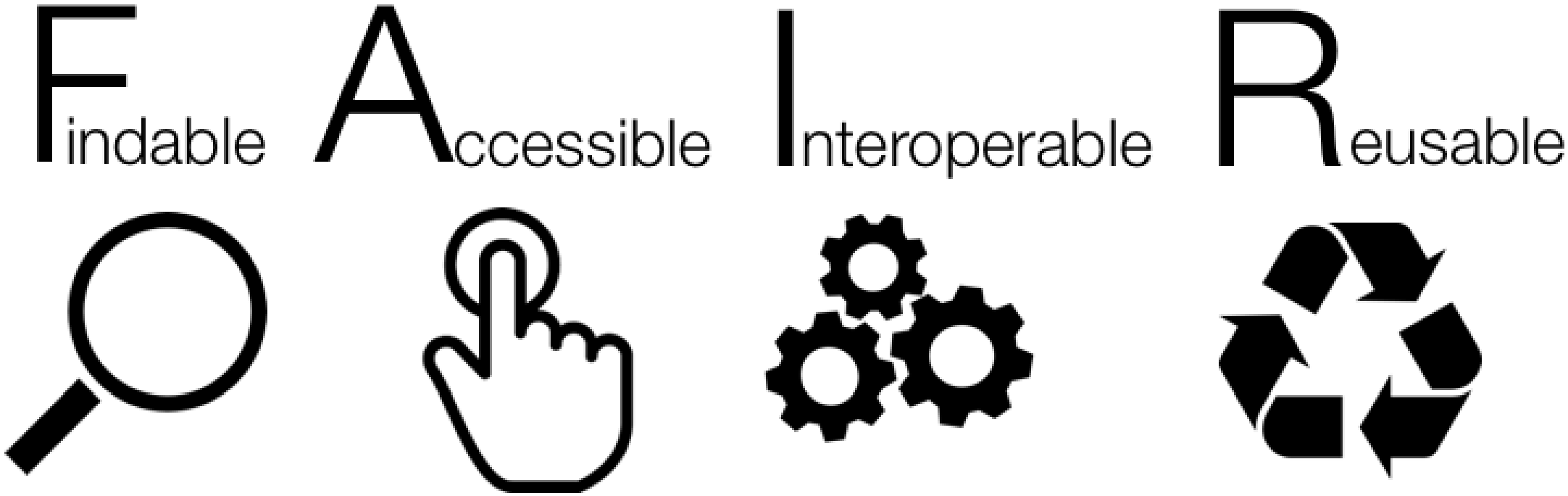


Links to non-existing group websites/databases.



Data / Code “on request”.

Data available but not reusable!



Wilkinson *et al.* The FAIR Guiding Principles for scientific data management and stewardship. *Sci Data* **3**, 160018 (2016). https://doi.org/10.1038/sdata.2016.18

FAIR principles



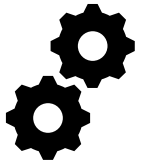
Findable:

- Data must be easily findable by humans and computers
- Datasets must be deposited in a data repository and have a persistent identifier (DOI)



Accessible:

- Data must be easily retrieved
- Metadata (data description) available even when the data are not



Interoperable:

- Allow integration with other data
- Data must be saved in open/common format (interpretable for various tools)



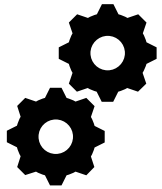
Re-usable:

- data should be well-described so that they can be replicated and/or combined in different settings
- conditions of the reuse should be stated with a clear license

How can you achieve FAIR data?



Reusability



Interoperability



Well-described, organised, and standardised datasets



Collecting FAIR data

Making **I**nteroperable and **R**eusable and data records

- ✓ Metadata
- ✓ File Naming
- ✓ Data Structure

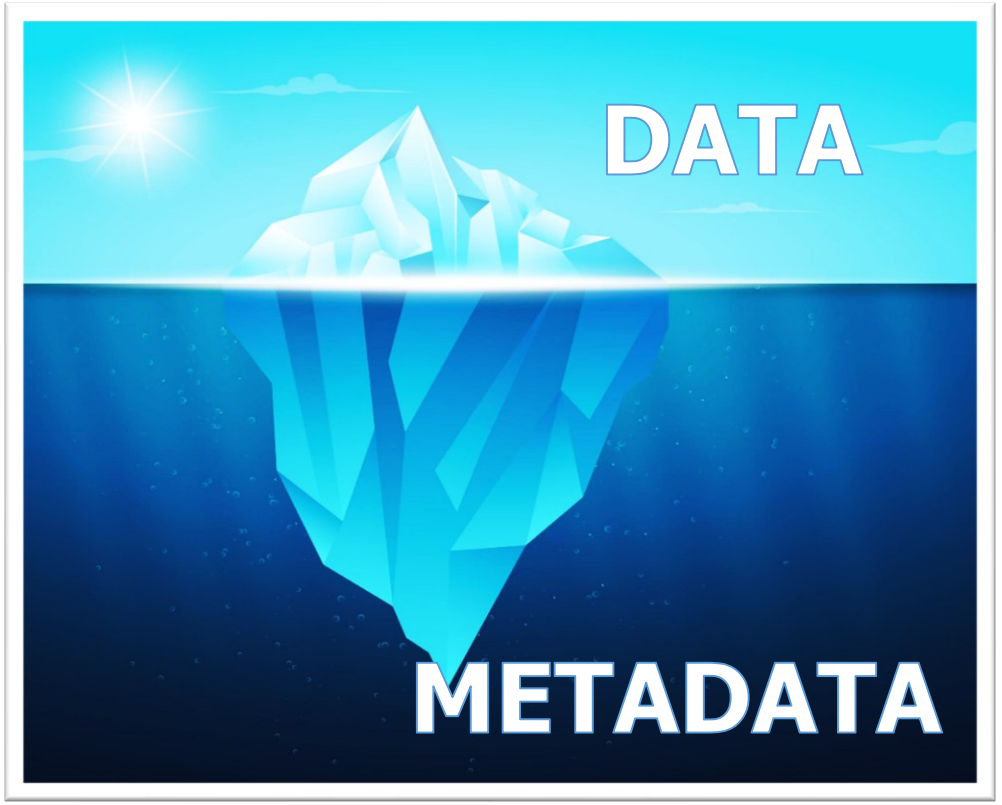
Metadata

What is metadata?

Metadata is the data about data!

Description/information of the data

The data has little or no meaning without metadata



"Iceberg image designed by pikisuperstar / Freepik"

✓ It should be continuously added to your research data (not just at the beginning or end of a project!)

✓ Metadata plays a significant role in making your data **FAIR.**

What information would you need to re-use this data?

Event code
MJTGQNM

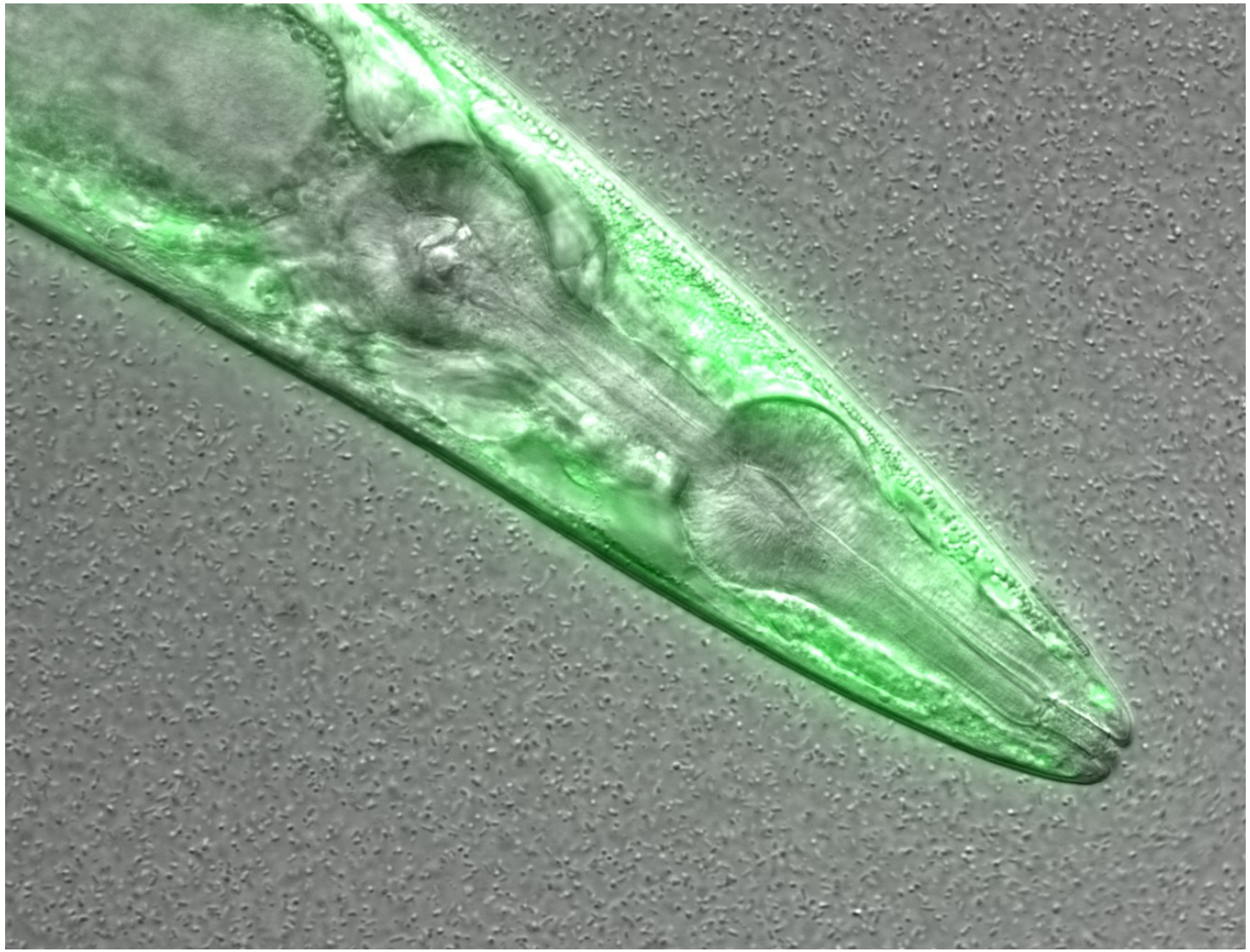


Figure credits: María Eugenia Goya

Metadata

Think as a consumer of your data, not the producer!

What information would another researcher need to understand or reproduce your data?

- Name of data creator/ Title
- Project purpose and experimental hypothesis
- Biological material, context, details on data acquisition, experimental conditions, etc.

+ Other datasets and publications associated with this image/ dataset (DOIs, links, etc).

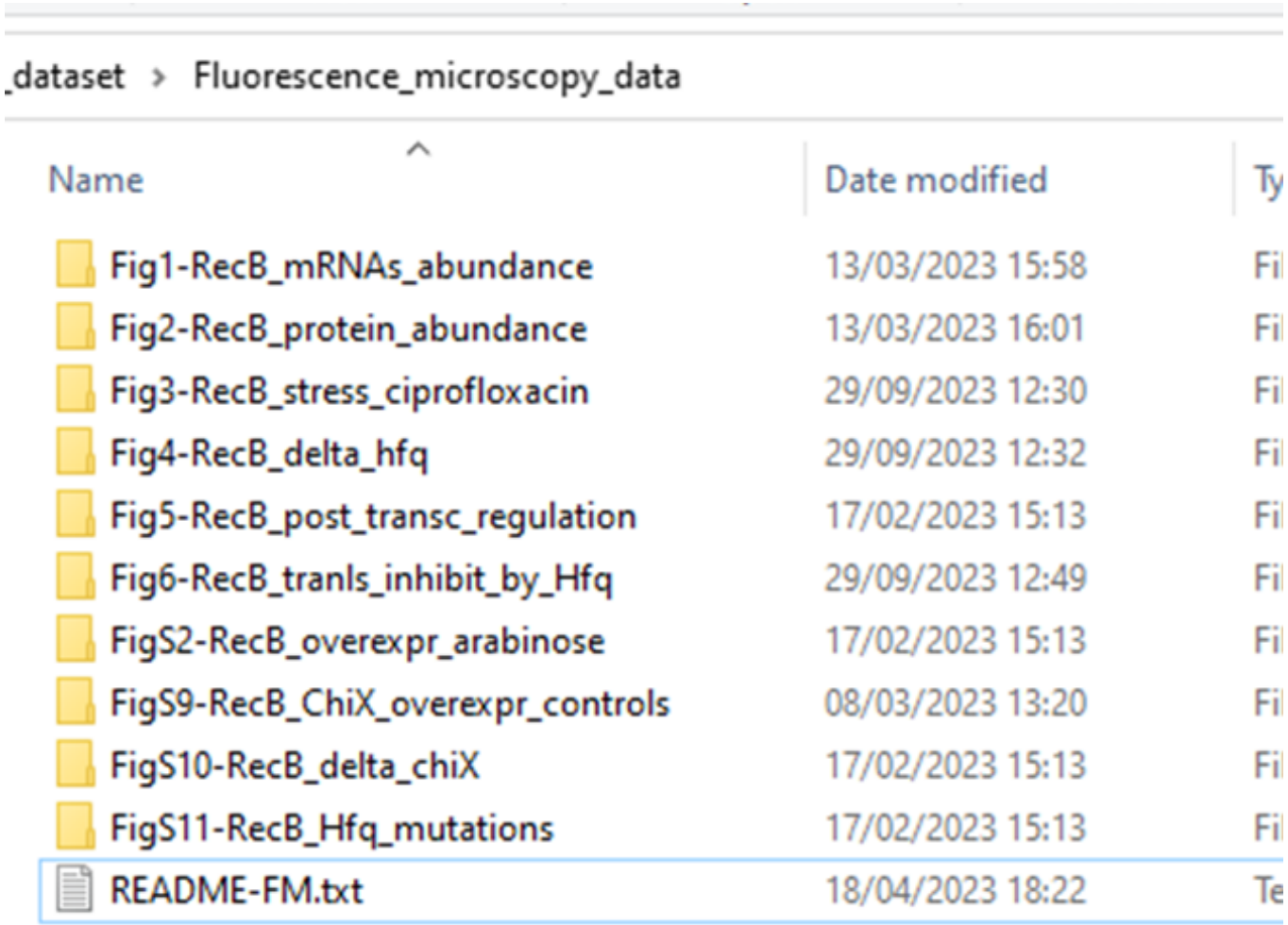
Metadata

	A	B	C	D	E	F
1	**Title**	Clock modulation of starch, pigments and nitrogen	Timet Project			
2	**Funder**	BBSRC-BB-V030557-1				
3	**Authors**	Sosa M	Pintos A			
4	**Contact**	Thee Bosss bio_rdm@ed.ac.uk				
5	**Growth protocol**	https://doi.org/10.21769/BioProtoc.126				
6	**Measurements protocols**	https://doi.org/10.1038/nprot.2006.232	https://doi.org/10.1038/nprot.2009.12			
7	**Study date range**	2019-12-05 to 2020-01-05				
8	**Cabinet**	Percival E-36L				
9	**Data description**	**Metabolites reported per g of fresh weight of 6-week-old plant leaf rosettes**				
10	**Data dictionary**					
11	**SD**	short days 6 h light				
12	**LD**	long days 18 h light				
13	**+SUC**	10 mg sucrose added per GM-agar				
14	**FW**	fresh weight				
15	**Kegg IDs**					
16	**Starch**	C00369				
17	**Sucrose**	C00089				
18	**Cholorophyll**	C01793				
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34						

metadata | data

Metadata in Excel

Metadata





Metadata in
README files

README files

README-FM.txt - Notepad

File Edit Format View Help

```
# Fluorescence microscopy data for "Fine-tuning of RecBCD expression by post-transcriptional regulation is required for optimal DNA repair in Escherichia coli"

doi: https://doi.org/10.1101/2021.10.23.465540

-----------------------------------------------

## GENERAL DESCRIPTION

This folder contains the fluorescence microscopy data presented in the aforementioned study, where the RecB  was detected either at the  mRNA level using single-molecule fluorescent in situ hybridisation (FISH) or at the prote
The data is divided in a "folder by figure" structure, meaning the data is separated according to the order that they appear in the publication figures (the prefix numbers in the folder titles correspond to the figure numbers

## FOLDER AND FILE NAMING CONVENTION:

The folder/file names within the figure folders are given as "genotype_technique_sample-number" or "treatment_technique_sample-number".
For example:
*"wild_type_FISH_S1" means cells from a wild type strain analyzed using FISH, sample 1.
*"delta_hfq_pQE80L_HaloTag_S1" means cells from delta hfq mutant harboring the backbone plasmid pQE80L analyzed using HaloTag, sample 1.
*"on_cipro_FISH_S3" means cells treated with ciprofloxacin analyzed using FISH, sample 3.
*"off_cipro_HaloTag_S2" means cells that were not treated with ciprofloxacin (control) analyzed using HaloTag, sample 2.
*"delta_recB_pIK02_arab_FISH" means cells from delta RecB mutant containing an arabinose-inducible plasmid (pIK02). Note that within this folder, there are "Arabinose_1pc", "Arabinose_10-1pc" etc, where pc means percentage.

Ciprofloxacin treatments, recB mRNA degradation with rifampicin and RecB removal rate experiments were performed in wild type strains.

## FOLDER CONTENTS

Each figure folder contains two main folders: "Images_raw" and "Quantification_results".
An "Example_masks" folder is also included in "Fig1-RecB_mRNAs_abundance" and "Fig2-RecB_protein_abundance". This contains examples of masks obtained for one FISH and one HaloTag experiment, respectively.
Negative control images necessary for setting the threshold in the spot quantification analysis are included for the same datasets of the example masks, if one wishes to re-run a FISH and/or HaloTag image analysis for training

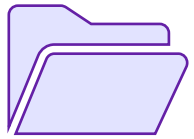
- "Images_raw":
The "Images_raw" folders contains subfolders with raw images in .TIF format obtained from the different experiments and samples, following the naming convention described above.
The images were obtained using bright-field and the mCherry fluorescence filter. The TIF files are clearly labeled as either "Brightfield" or "mCherry" in their names. The naming convention of TIF files does not correspond to
The last Z-stacks of bright-field images were used as were used as an input for cell segmentation and production of cell masks. In some instances, extra bright-field images were manually obtained refocusing the microscope as t
Negative control images were used to set up the threshold in the image analysis workflow (setup of the initial parameters in InitializeSpotRecognitionParameters.mat and TestSpotThreshold.mat. TestSpotThreshold.mat may be neede
The examples of negative control images for a FISH experiment are in "delta_recB_FISH_S2", and in "wild_type_no_HaloTag_neg_ctrl" for HaloTag.

- "Quantification_results"
The "Quantification_results" folder contains numerical data in .CSV format. These data was extracted from matlab data files generated by the image analysis pipeline.
During the export process to .CSV, rows containing NaN values were removed (NaN values appeared when the intensity of the total spot intensity of the cell could not be estimated) and descriptive header fields were added.
The rows correspond to each segmented cell.
The data from FISH experiments contain 9 columns, while the data from HaloTag experiments contain 8 columns.
```

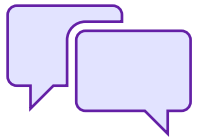
File Naming

File Naming

Naming + Organisation Goals



Find files quickly



Ability to tell the file content without opening it

Standardised Naming Conventions are…

- ✓ Consistent.
- ✓ Encoding of experimental factors.
- ✓ Meaningful to you and your collaborators.
- ✓ Give you a sense of the content.
- ✓ Easily identify if something is missing.

File Naming

File names include:

- Date → ISO format: YYYY-MM-DD or YYYYMMDD (e.g. 20241106)



Add the date at the end of the file *UNLESS* you organise files chronologically

- Version Number → Use leading zeroes (e.g. v005 instead of v5)
- Prefix for Numerical Value → e.g. S003, TR005

File Naming

File names include:

- Relevant Biological or Experimental Info

e.g.:

✓ Species name, Genotypes, Site, Patient ID, Drug treatments/ Doses, Environmental conditions, Markers, Technique.

✓ Data operation (normalized, cleaned, detrended, clustered)



File Naming

File names include:

- File Format Extension → e.g.: .docx, .xlsx, .mov, .tiff



https://dragondatarecovery.co.uk/what-are-file-types-and-how-to-identify-them/

File Naming

An excellent example:

LD_phyA_off_t04_2020-08-12_norm.xlsx

We know that the file is…

- from an experiment long day (LD).
- for genotype phyA.
- with media without sucrose (off).
- at timepoint 4 (t04).
- normalised data (norm).



File Naming

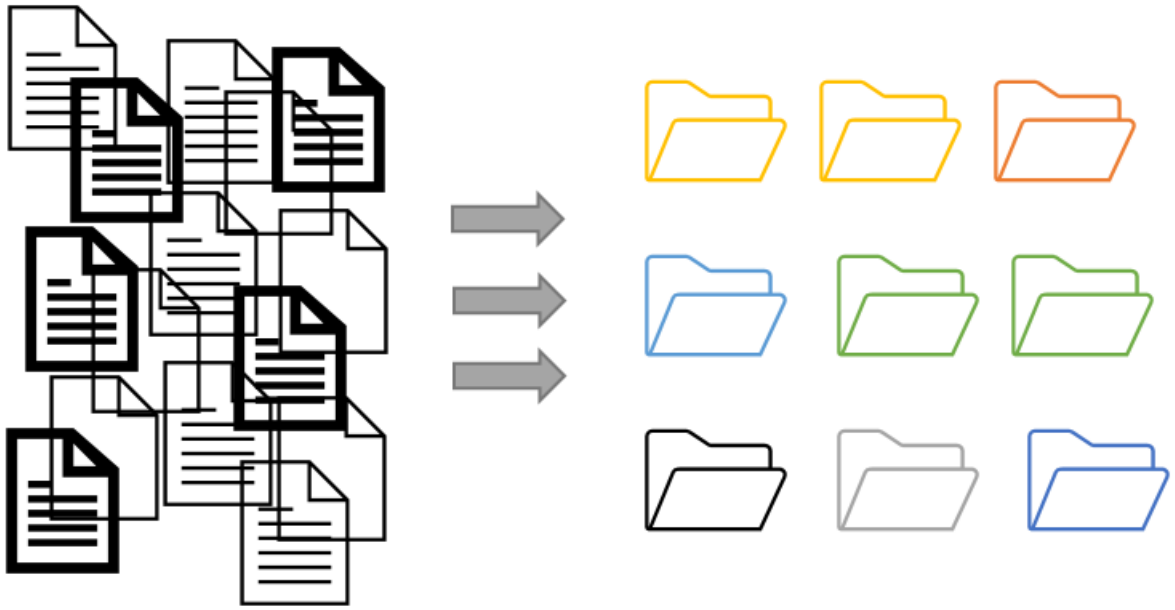
Best to avoid the following: 

- Spaces → *use _ or - instead*
- Commas + Special Characters → *e.g. ~ ! @ # $ % < > ?[ ] { } '*
- Language-specific characters → *e.g. ó ę ż ï ň ũ*
- Long names → *fewer than 30 characters*
- Deep Paths with Long Names → *i.e. deeply nested folders with long names*
- Parent Element Info →

ELN_MI_20200101.tiff in Electron_Microscopy folder

Data Structure

Project organization



messy/unstructured

organised folder structure

Figure credits: Andrés Romanowski

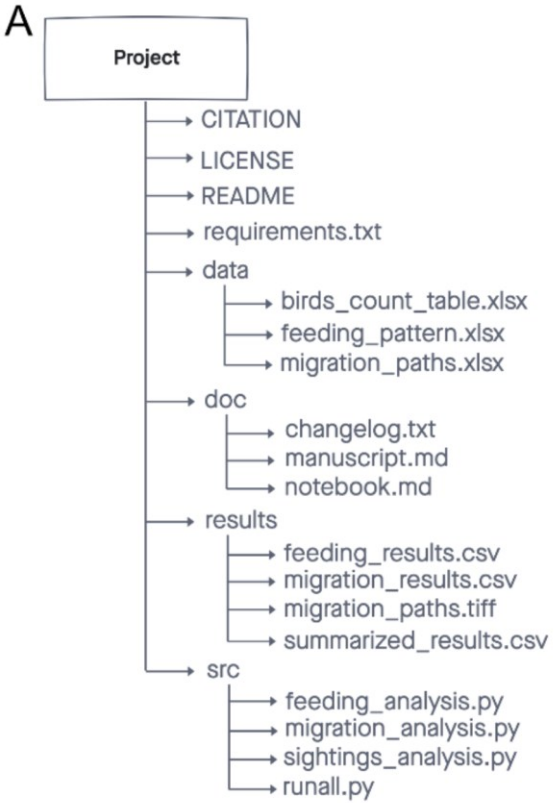
Folders vs Files

- Folders permit grouping relevant data together
- Folders help to keep files names short

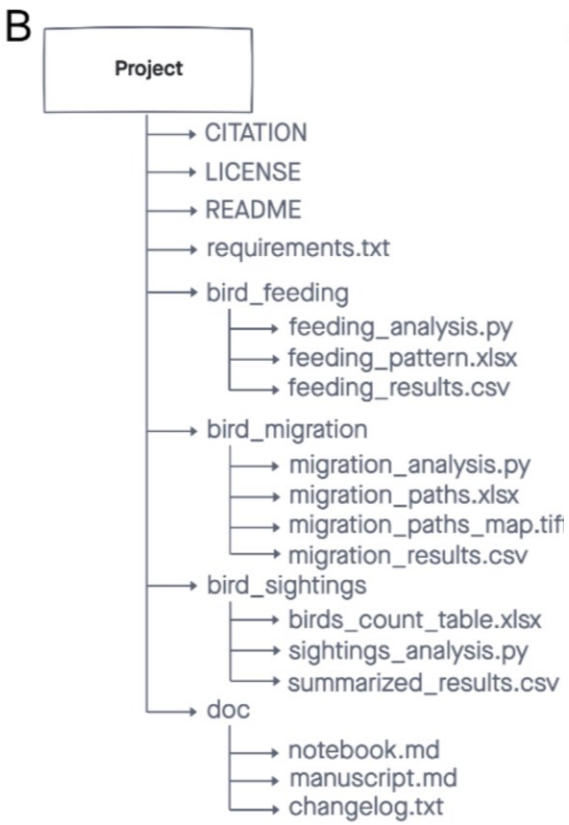
Project organization

Event code
SKJNAKU

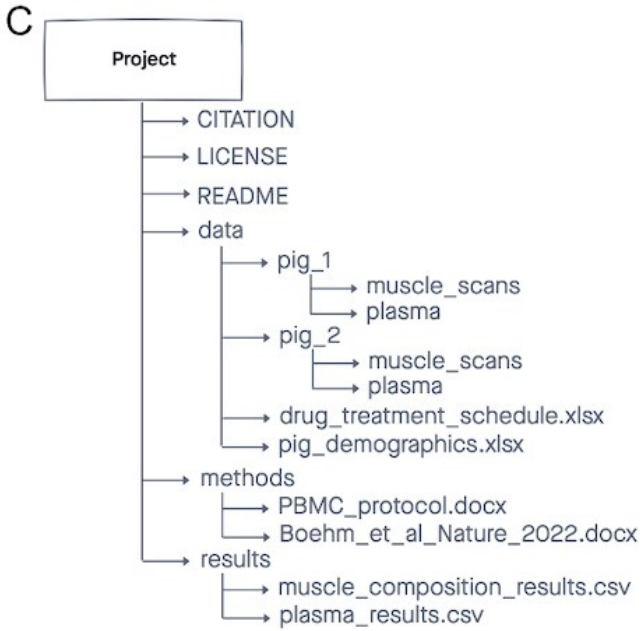
A



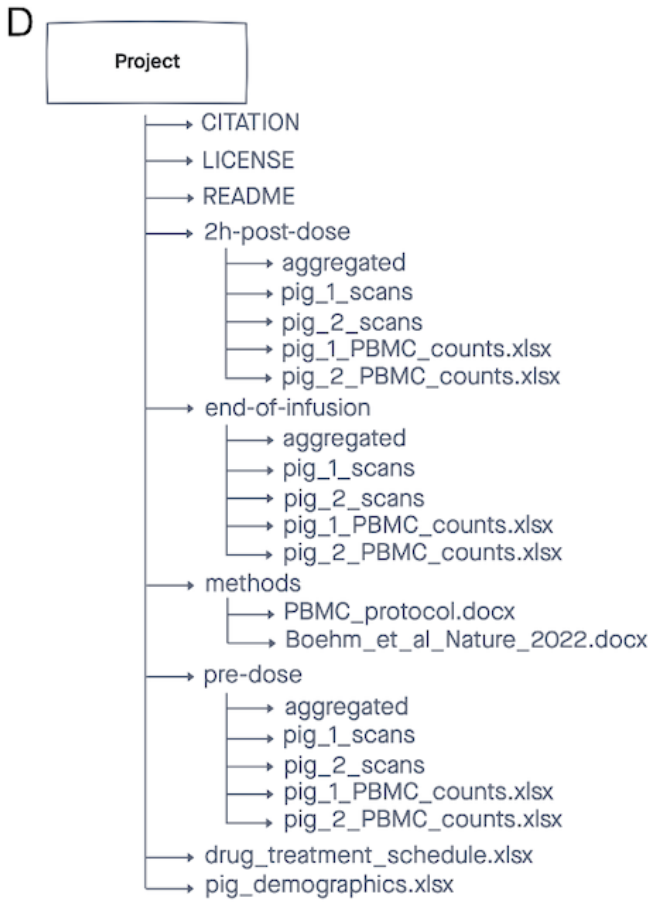
B



C

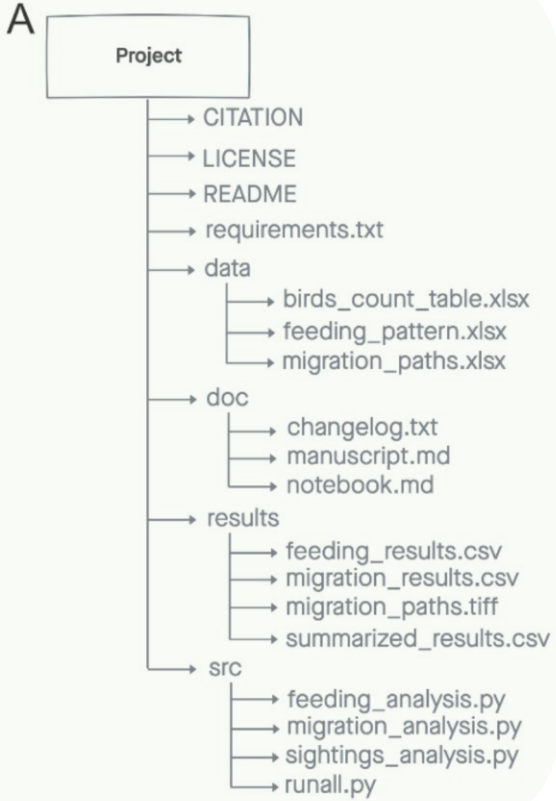


D



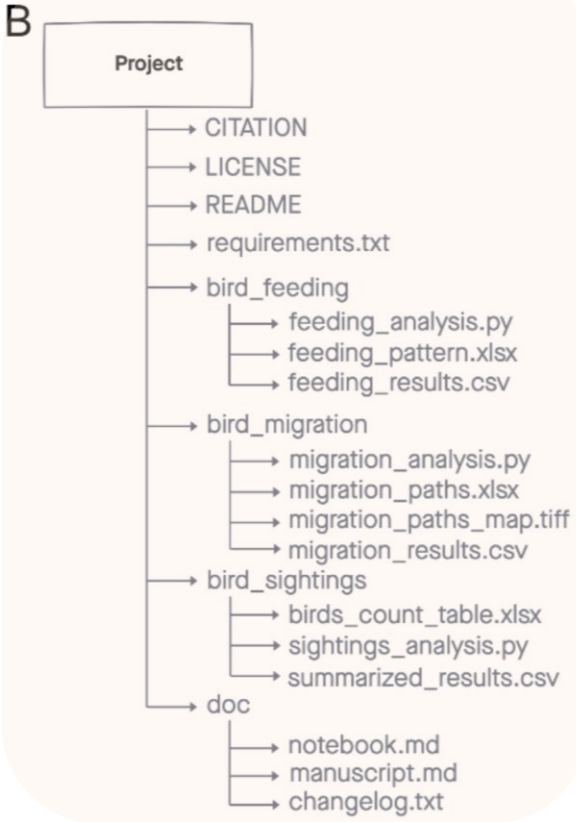
Project organization

A



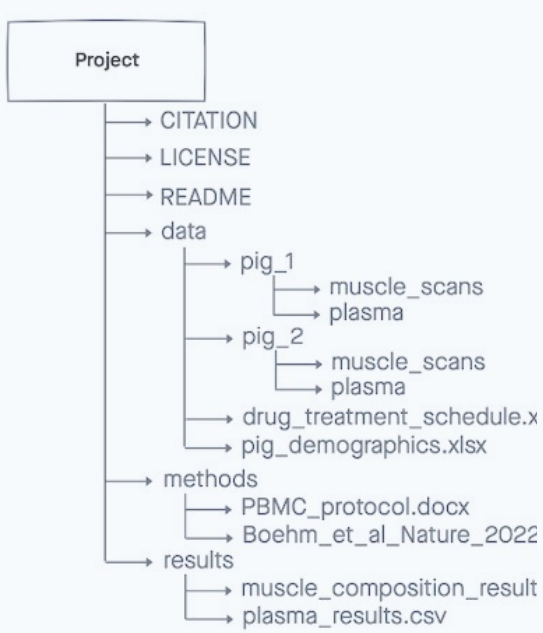
[Good enough practice in scientific computing]

B



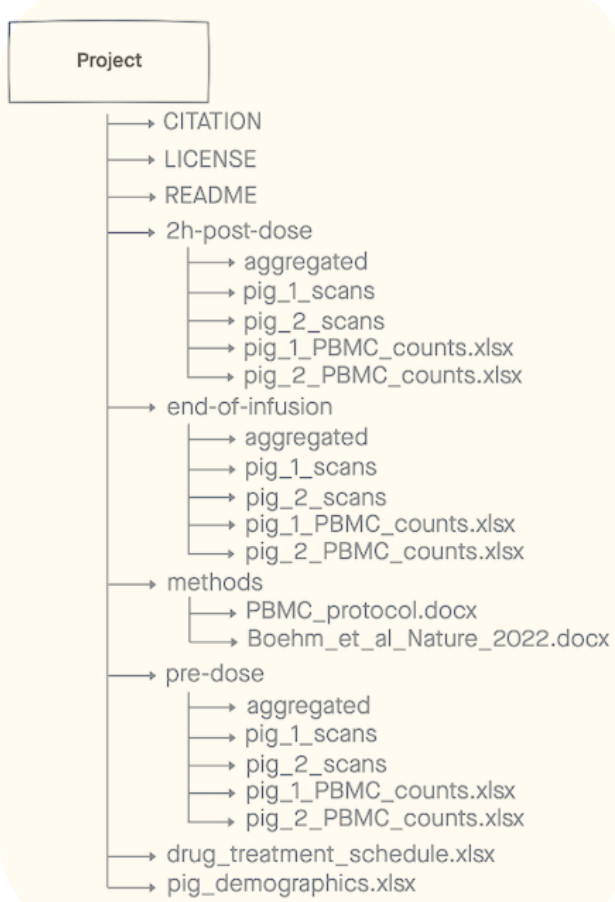
Organise by analysis or by figure

C



[Brain imaging data structure]

D



Outcomes of experimental conditions

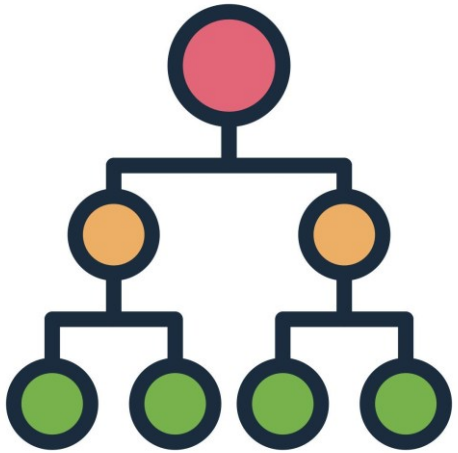
Record your data structure in a README

Add to your dataset directory

- PROJECT_STRUCTURE (README) file → naming convention, directory structure and abbreviations



Metadata in README files



Exercise #1

- **Resource**: Trainee's Personal Data Files or Provided Data Files
  - Exercise materials

- **Task A**: Edit file names and folder structure according to best practices
- **Task B**: Edit the ReadMe file with the following info:
  - Folder description
  - Author(s) info
  - Description of folder contents
  - Naming conventions used



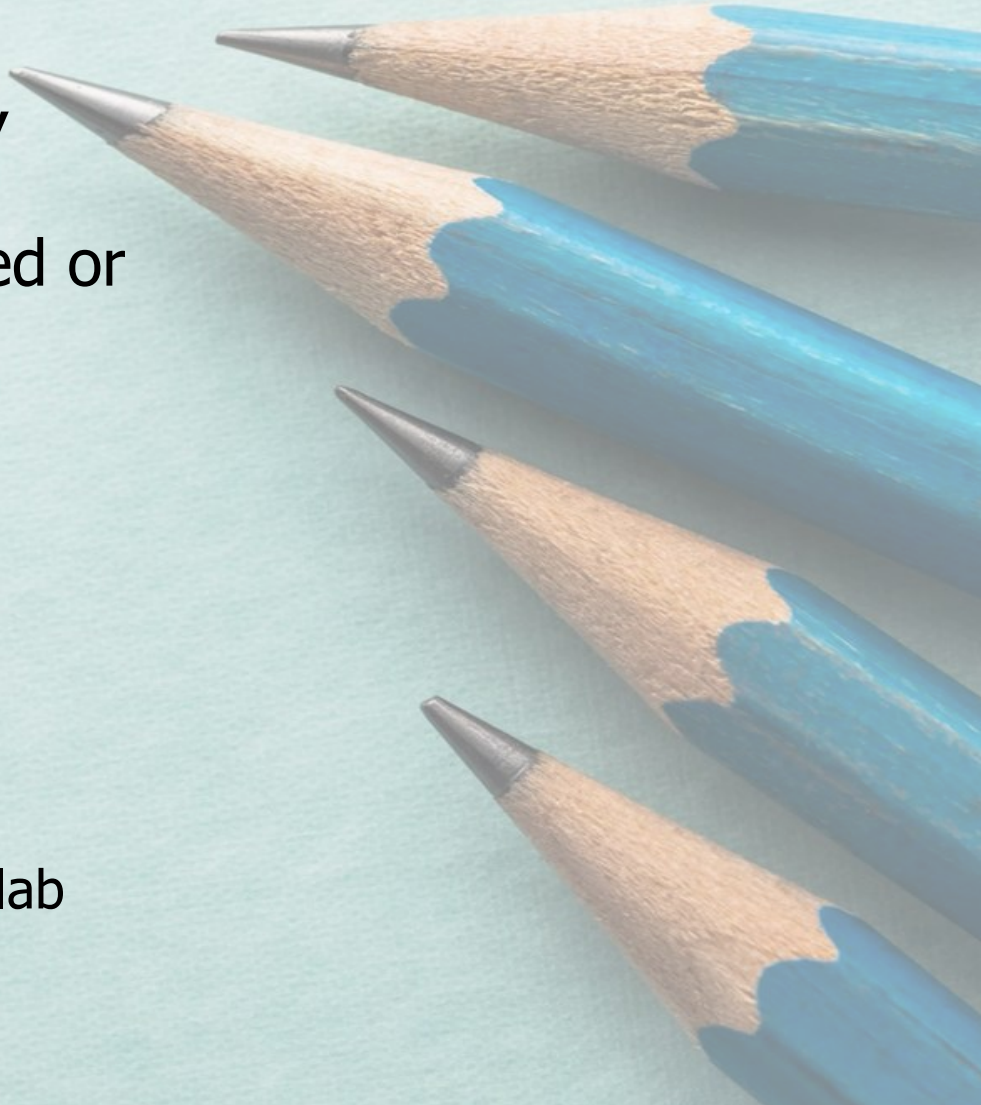
Meaningful_name_date.extension

- **Output**: Basic folder stub

Enduring Understanding

Finding your data and is impacted by file naming, metadata, and folder structure.

(Meta)data that cannot be found cannot be reused or shared



Practical Applications

- Understand your own work in 2 years time!
- Interlinking data analysis results to original raw datasets
- Make it easier to pass on your work when you leave the lab
- Make your data sharing-ready

Please confirm your attendance here (for Graduate school):





Break 1
(10 min)



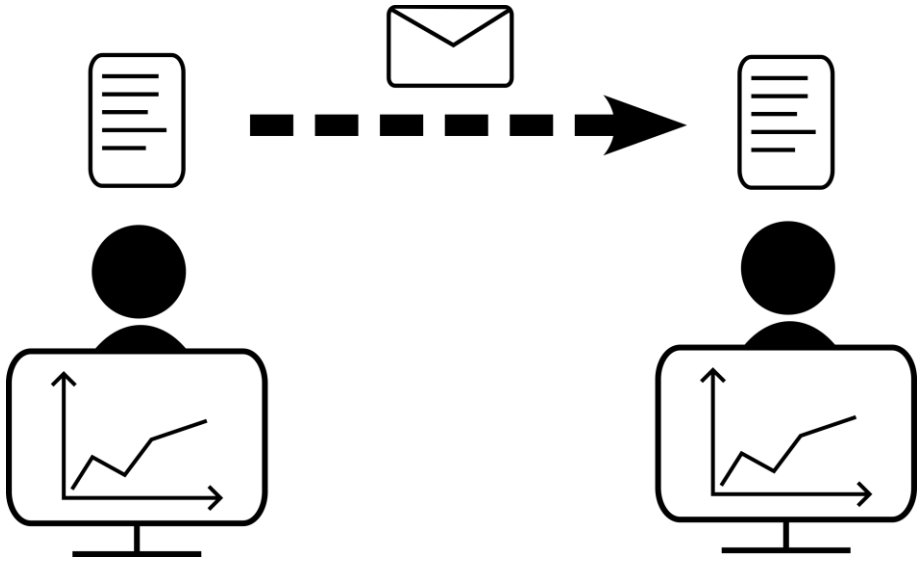
Managing **INTEROPERABLE** Data

In what ways and why is data interoperability arguably the most complex aspect of FAIR practice?

- ✓ File Formatting + Types

What is data interoperability?

The ability to exchange and use data across multiple systems



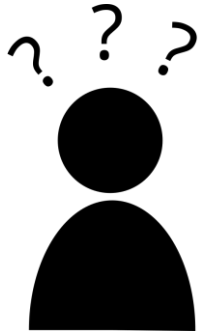
The ability of diverse datasets to be merged and aggregated in meaningful ways

- 0.5L milk
- 3 eggs
- 150g flour

+

- 1 gallon milk
- 200g egg
- 3oz flour

=



File Types and Formatting

FAIR file formats in biological practice

1

Use .csv or .xls files for numerical data. **Never** share data tables as word or pdf

2

Provide underlying numerical data for all plots and graphs

3

Use common file formats (domain specific)

4

Convert proprietary binary formats to open ones. For example convert Snapgene to Genbank/SBOL, microscopy multistack images to OME-TIFF

Data formatting

- **Use standard formats wherever possible**
  - Dates in ISO8601 format: 2025-10-16 14:53:02
- **Be consistent!**
  - For a variable with two levels: TRUE/FALSE, 0/1, Yes/No,…?
  - Use the same units (often standard within a field)
- **Think ahead**
  - Every time you change your data formatting, you will either break interoperability with previous data, or generate work to re-format it

Exercise #2 (10 min)

Resource: File examples

- Exercise materials

Task A: Which of these file formats are interoperable or not?

- Try to copy the "Supplementary_table.pdf" data into Excel
- Try to open "SI_fig_1.opj"
- Is "Fitting_model.png" interoperable?
- Is the example Python script interoperable?
- Open the "Ens_data.xlsx" file: is it interoperable?

Task B: What alternative file formats would increase interoperability?

Output: Context about the interoperability of different file formats



Enduring Understanding

At any scale, data interoperability can be logistically challenging (i.e. between different software) and conceptually challenging (i.e. removing data from its original context may yield different findings).



Practical Applications

- Let other people reuse your data (and cite you for it!)
- Facilitate collaborations
- Keep access to your data if you lose access to software



Managing FAIR data records

Using digital tools to make data **FAIR**

- ✓ Electronic Notebooks
- ✓ Jupyter Notebooks
- ✓ Repositories: making your data **F**indable and **A**ccessible!

Electronic Lab Notebooks

Good Lab Records

Laboratory Notebooks

Contain all relevant details (what, who, when, why and how)

Which project(s) is the record part of

Information on lot/batch numbers

What happened and what did **not** happen (data, including images)

How you processed and analysed the data

Your interpretation (and the interpretations of others if important) and next steps in the project based on these results

Should be well organised for ease of navigation (indexed, labelled, catalogued)

Accurate and complete: include all original data and important study details (metadata).

REMEMBER!

Both protocols and laboratory records need to be detailed, accurate and complete.

Good Lab Records

Protocols

Give credit to the person who created the protocol, if it was not you.

Complete and detailed instructions describing why and how to do an experiment

What special materials and instruments are being used and where they were obtained

Health and safety advice and how to dispose of waste

Allow repetition of your procedures and studies by yourself and others



Figure credits @sketchscience

ELNs

Let's make ourselves a cup of tea.

Here's a peer-reviewed protocol for making tea

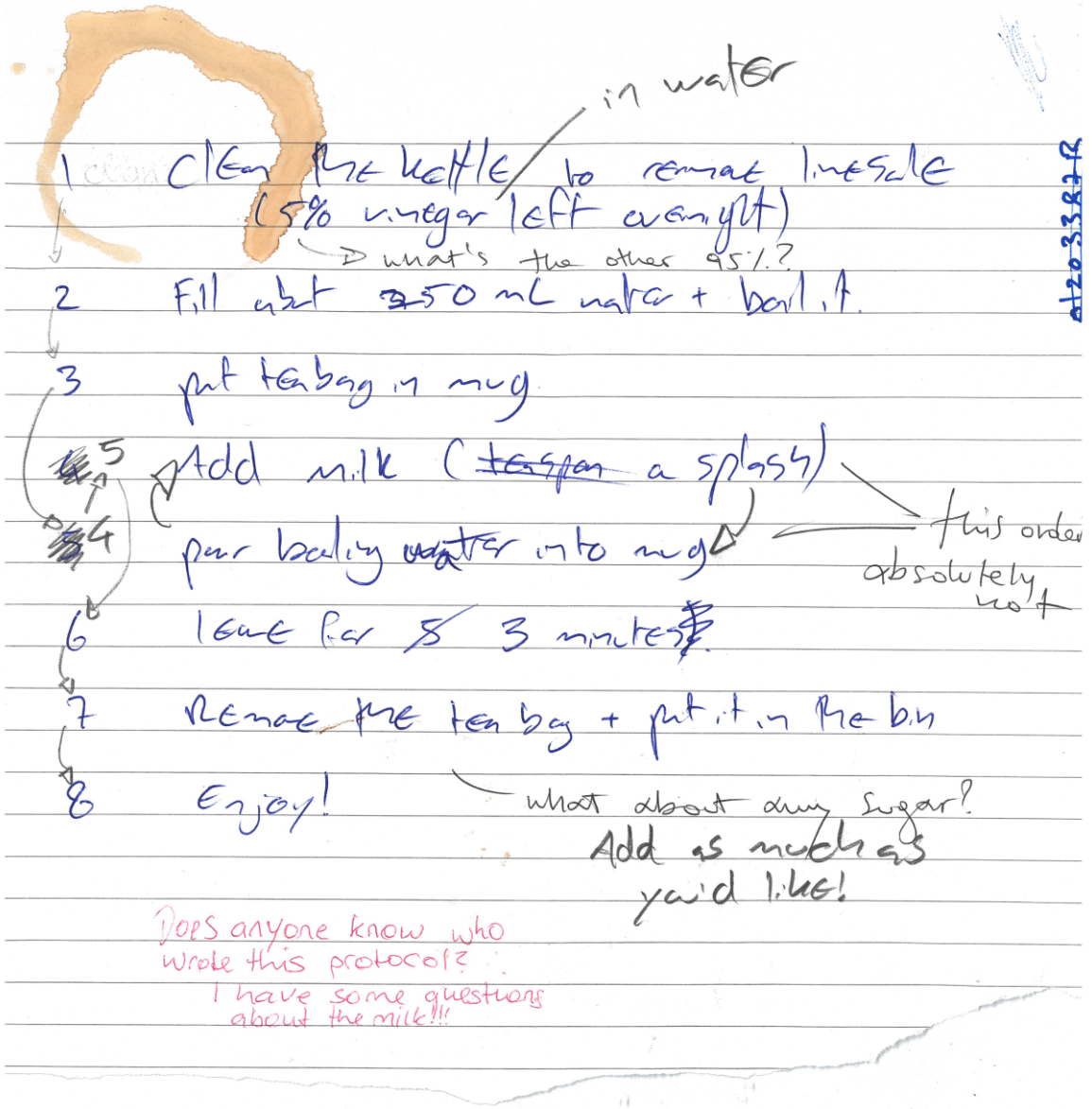
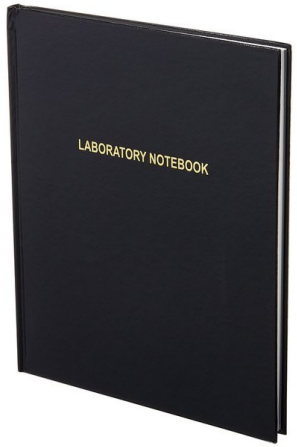


Figure credits: Dr Ines Boehm and Ben Thomas

Paper



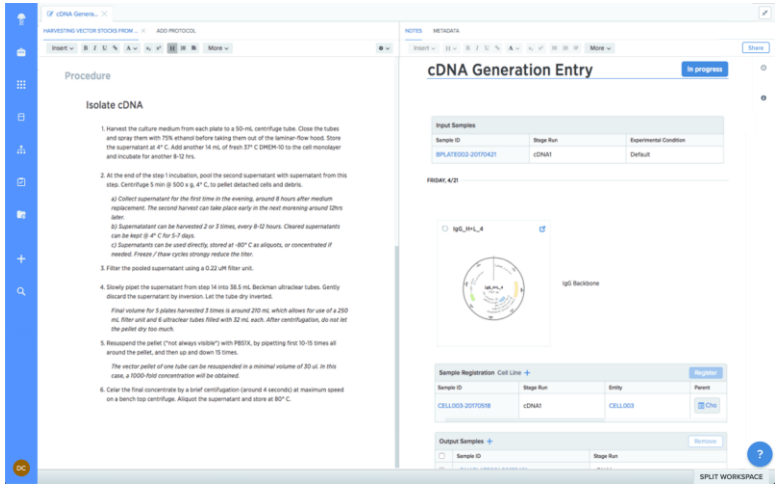
- It does not require the internet.
- Directly draw on your records.

Pros

- Can be lost or damaged
- Handwriting is not always intelligible
- Hard to edit and share
- Hard to reuse information
- Not findable/ accessible

Cons

ELN



- Your data is organised and searchable.
- Allows off-site archiving.
- You can have templates (e.g.: protocols/assay / entries) - Easy to copy data entries
- Can be shared instantly; global access within a group.
- Doesn't take up physical space (no record rooms/folders)
- Regular backups.

Pros

- Requires internet access
- Better if you have a laptop or tablet

Cons

How do I choose an ELN?

Does my group already use one?

Yes ↓

Use that one!

↓ **No**

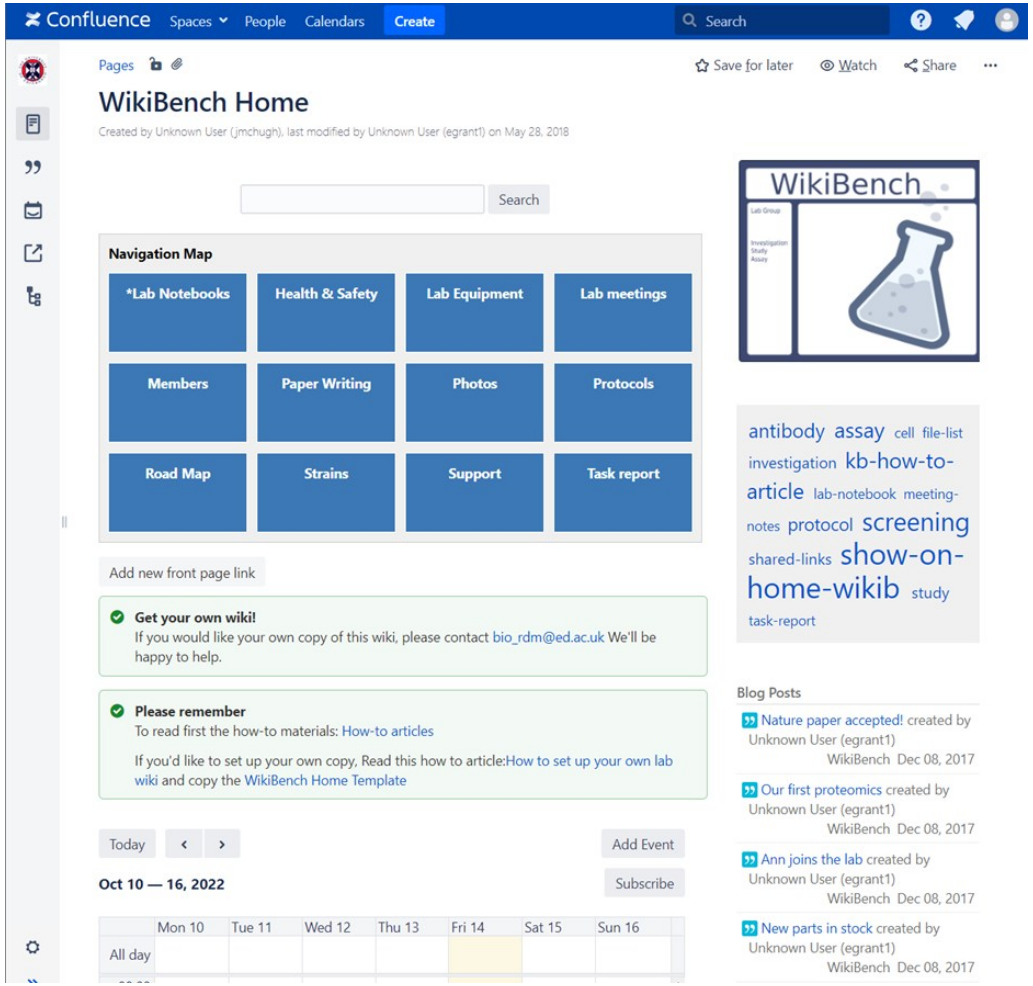
Things to consider:

- Cost
- Traceable? (data cannot be deleted and is time-stamped)
- Where are the servers located?
- Can the data be downloaded?
- What happens to the data when someone leaves the group?



Which ELNs are used in SBS?

WikiBench (UoE Wiki)



Benchling

RSpace

labarchives
Better Science

jupyter

For code

protocols.io

For keeping/sharing protocols

Example of good record keeping



Aims of the experiment



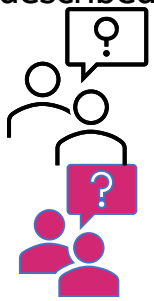
Samples are described



Protocols are written down and referenced



Result figures and description



What worked and what did not; potential solutions

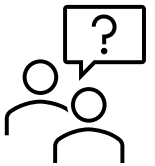


Final conclusions

Issues accessing the link? Please let us know ☺

Benchling

https://benchling.com/s/etr-0FdV1H0rpWeHk4H72NOg/edit



ELNs Resources from BioRDM

https://www.wiki.ed.ac.uk/x/f0SkGw

Hands-on tutorial on Benchling:

https://www.wiki.ed.ac.uk/spaces/RDMS/pages/463750251/Benchling+quick+tutorial

Benchling

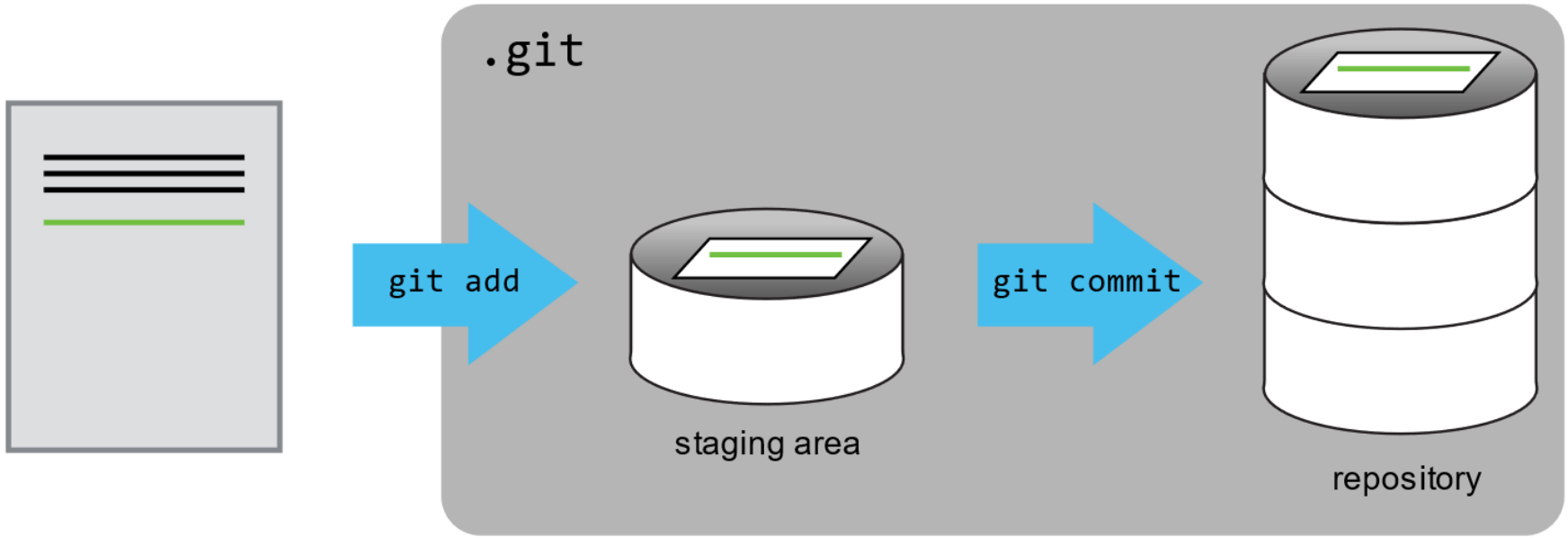
bio_rdm@ed.ac.uk

Demo



[Demo materials]

Version control



software carpentry

If you are producing code, you must use version control Git

https://swcarpentry.github.io/git-novice/

git

Exercise #3 (15 min)

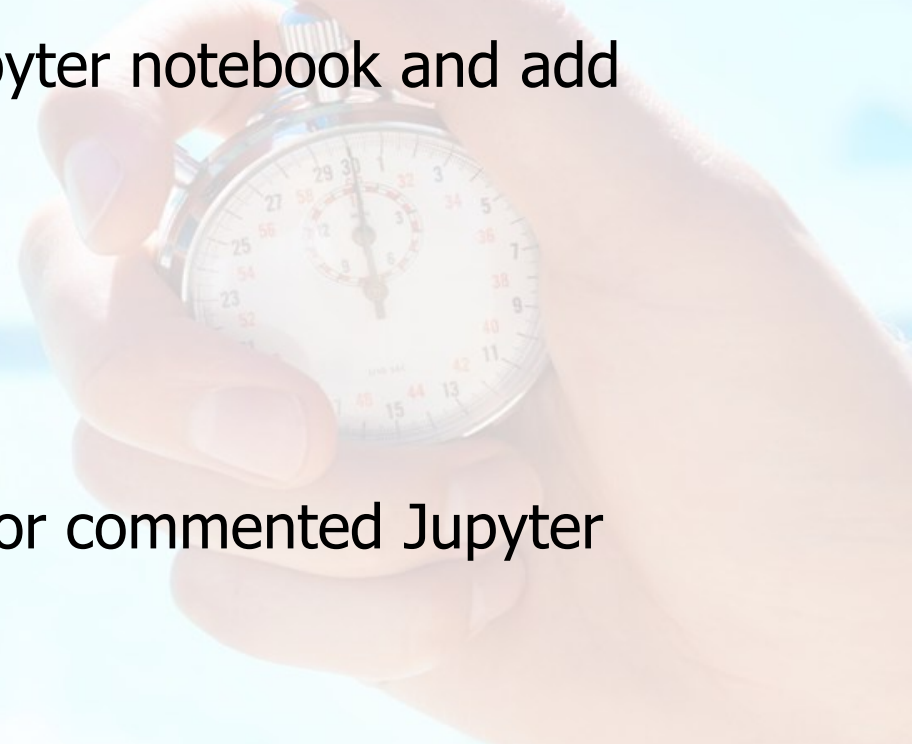
Resource: Wet lab protocol or Piece of Code
- Exercise materials

Task Option A: Create a protocol on an ELN
- benchling.com

Task Option B: Transform a piece of code into a Jupyter notebook and add comments to it
- https://mango.bio.ed.ac.uk/jupyter
- FAIR2025

- https://jupyter.org/try-jupyter/tree/

Outputs: Registered ELN account and protocol stub or commented Jupyter Notebook



Repositories

Repositories



Findable



Accessible

- Deposit data to an external, reputable public repository.
- Repositories provide **persistent identifiers** (PIDs), catalogue options, advanced metadata searching, and download statistics.
- Uploading data to a repository **requires rich metadata**

Repositories



Some repositories can also host private data or provide embargo periods, meaning access to all data can be delayed.



There are general “data agnostic” repositories, for example: Dryad, Zenodo, FigShare and UoE’s own DataShare .



Or domain specific, for example: UniProt – protein data, GenBank – sequence data, MetaboLights – metabolomics data.

Exercise #4 (5 min)

Event code
JXFENEO

Resource: https://zenodo.org/records/6349579

Task: Identify the elements that make a dataset record FAIR in the Zenodo record.

Outputs: Practical knowledge about the FAIR elements in a public repository.

Enduring Understanding

Digital tools like ELNs are imperative for data reusability and to boost collaborative work.

Depositing data in a repository will make your data **findable** and **accessible**, but you need to describe/organize/ format it appropriately to make it **interoperable** and **reusable**



Practical Applications

- Facilitate work within your team
- Keep reliable records of your work
- Find information quickly
- Share FAIR data

Please confirm your attendance here (for Graduate school):





Break 2
(10 min)



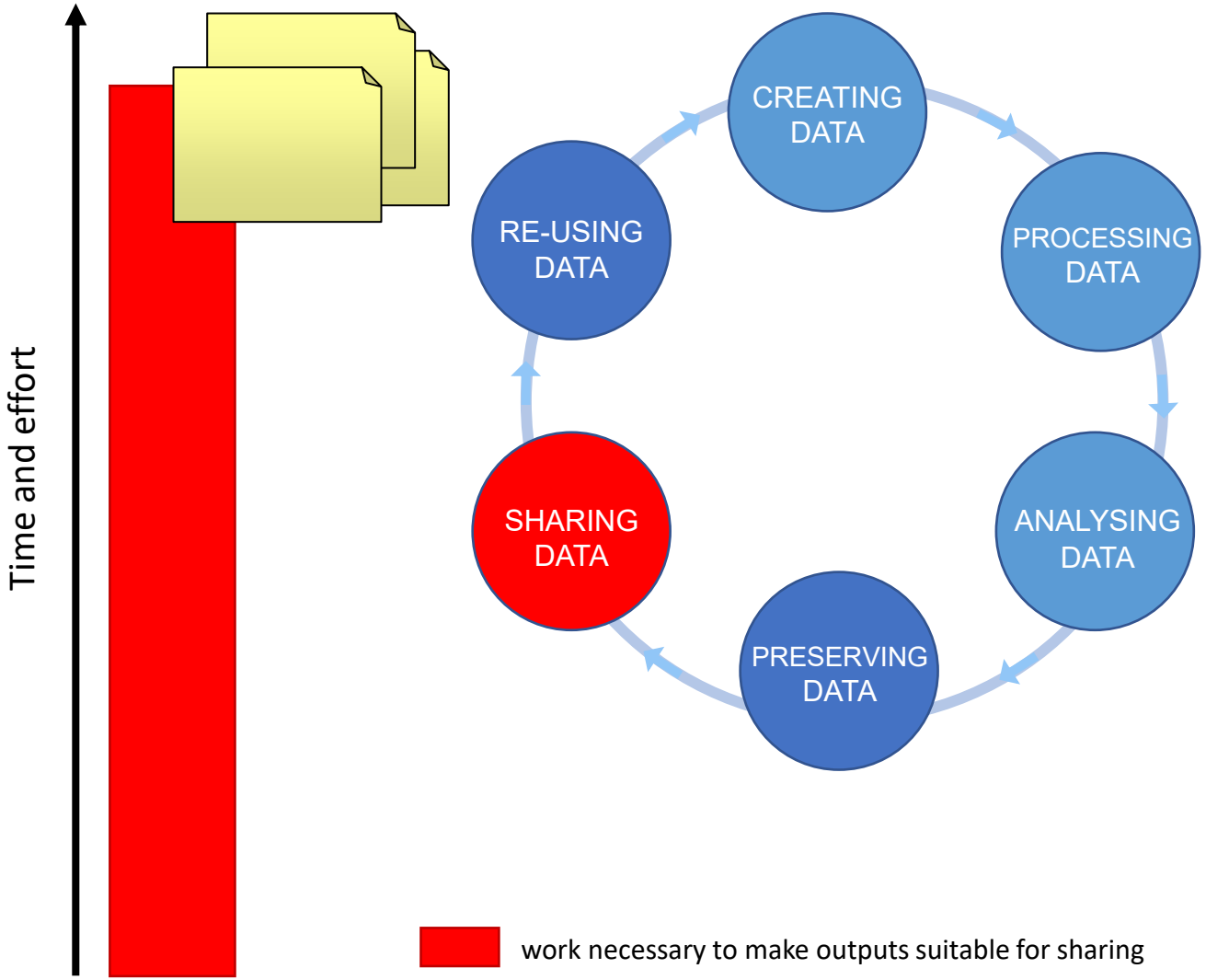
Planning ahead

To what extent is reusable data consequential and for whom?

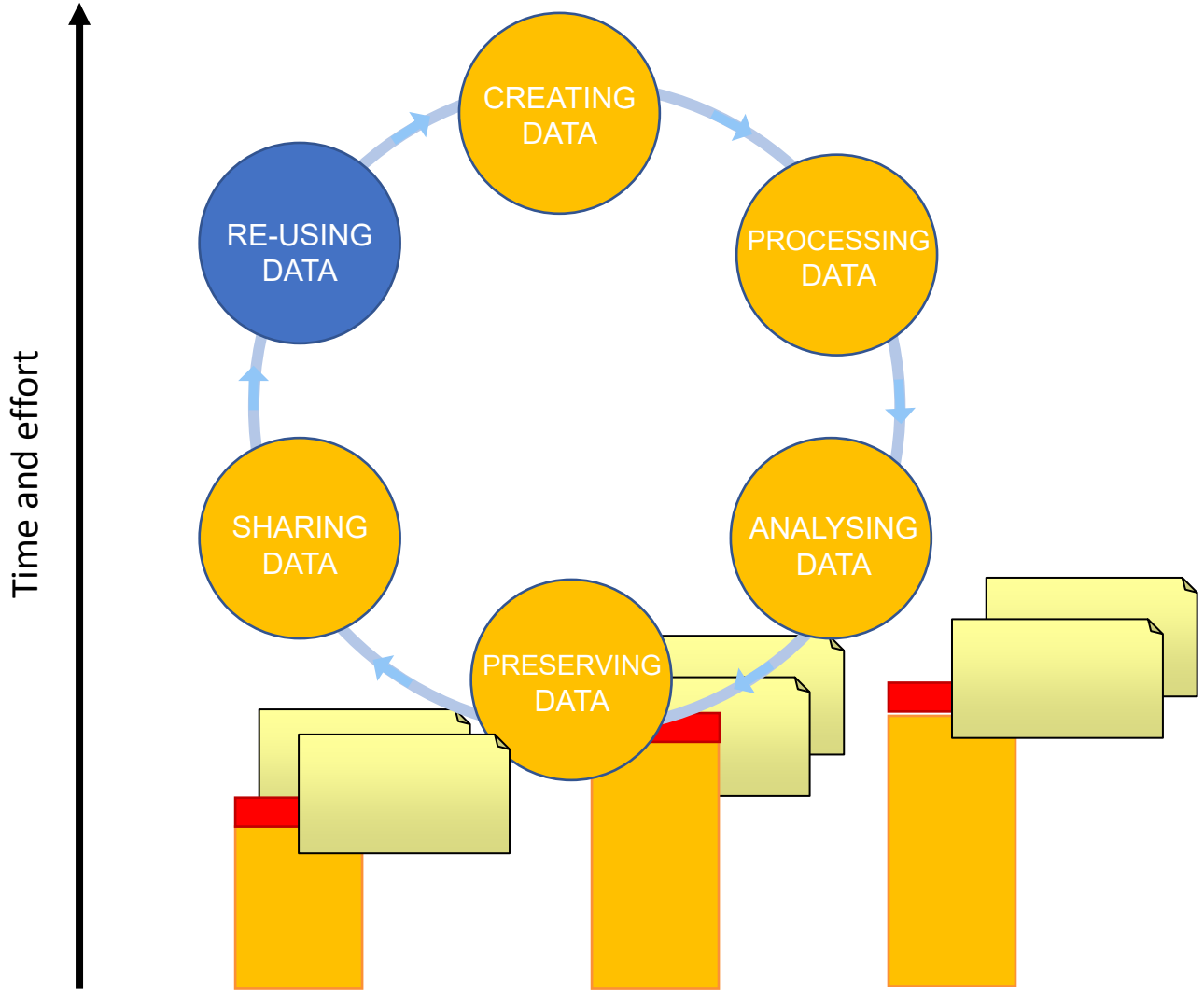
- ✓ Data Sharing Cycle
- ✓ Data Management Plans
- ✓ Research Outputs

Data Sharing Cycle

Sharing as part of the workflow



Sharing as part of the workflow



Plan ahead

Data management plans (DMPs)

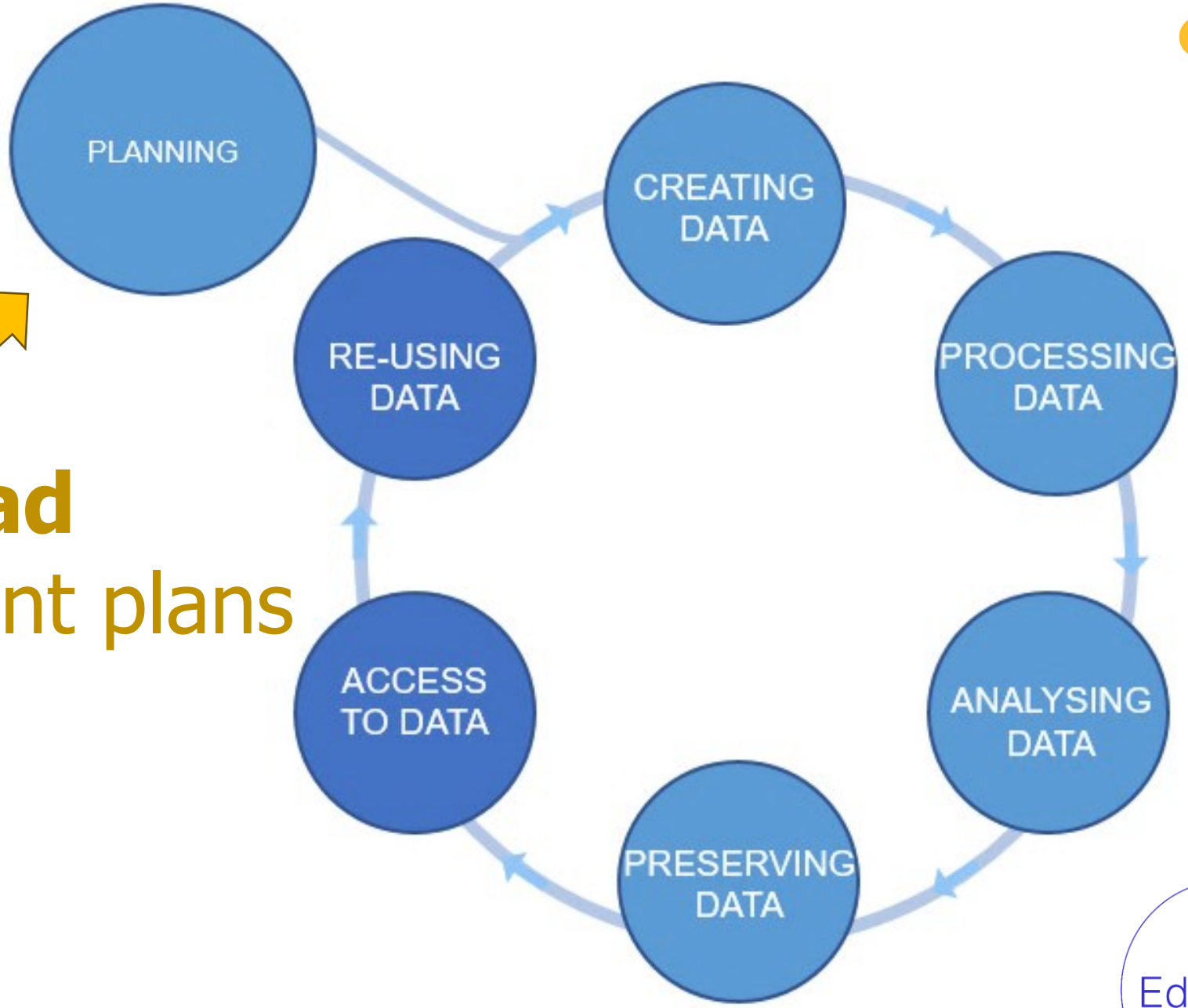


Figure credits: Tomasz Zieliński and Andrés Romanowski

Ed-DaSH

Planning ahead

Data Management Plans

Data Management Planning: What? How? Where?

What: thinking about what data you will generate.

✓Types of data, format, and volume of the data

How: thinking about how you will:

✓Organise, describe, securely store, and back up your data.

Where: to ensure your data is available for replication and reuse.

✓Archiving, publishing, and conditions of re-use.

Writing a DMP for your Research

Tools and courses

- DMPOnline https://dmponline.ed.ac.uk

Digital Research Services (RDS) Hands-on workshop(s)

- Understand the necessity/benefits of producing a DMP.
- Discover how to register for and use DMPonline.
- Draft a basic DMP!



https://library.ed.ac.uk/research-support/research-data-service/research-data-training-skills/workshops-courses



Planning ahead

BioRDM DMP advice

https://www.wiki.ed.ac.uk/display/RDMS/Data+Management+Plans

Exercise #5

Resource: PhD project background info

Task A: Answer or reflect on the following questions on WooClap

- What data types do you expect to generate?
- What file formats will you work with?
- How much data do you expect to generate over your whole PhD?
- Where will you store the data?
- How will you transfer data (e.g. to collaborators)?
- What folder structure will you use for your projects?
- Will the data be made publicly available at the end of your project?
  - If yes: on which repository could you make the data available?

Event code
HQAMHJP

Output: Preliminary Data Management Plan

Planning ahead

Research Outputs

Your musts

According to policies (the School, the University and the funders):

You are responsible for the safety of your data (both storage and access)

- Back up (almost) everything created by a human or recorded by a machine as soon as it is created.
- Physical HDs only are a failure-prone option and are not a valid solution for backup of scientific data
- Always backup your files in 3 places, at least one should be off-site.
- A robust backup cannot be achieved manually – **the University gives you DataStore**

Your musts

According to policies (the School, the University and the funders):

- You need a DMP for your project
- You have to publish Open Access
- You have to make available all the data underlying publications



shutterstock.com · 2087491684

Writing a DMP for your Research

Annual Review: You must write a basic DMP for your project

Data Management and record keeping

Following your Data Management training, you will be familiar with the importance of Data Management Plans. You should also have discussed record keeping best practices with your supervisor(s). Use this section to describe your plans and provide an update on progress/changes. Questions are provided to help guide you. See also: https://www.wiki.ed.ac.uk/spaces/RDMS/pages/347997313/Data+Management+Plans

Data Management Plans:

- *What data will you generate? (Data types, file formats, amount of data)*
- *How will you work with it? (How it will be organised, stored, backed up)*
- *Where will it be available later? (Publishing, archiving and permissions for re-use)*

Record keeping:

- *What type of lab notebooks are being used? Are they maintained adequately?*
- *If relevant, how are reagents and strains catalogued?*

For your narrative CVs…

… you need to have as many public outputs as possible:

- Datasets in public repositories
- Public protocols
- Code packages
- Public engagement, industry
- Involvement in committee work, student groups etc.

(BioPod, Doors Open Day, Festivals, etc)



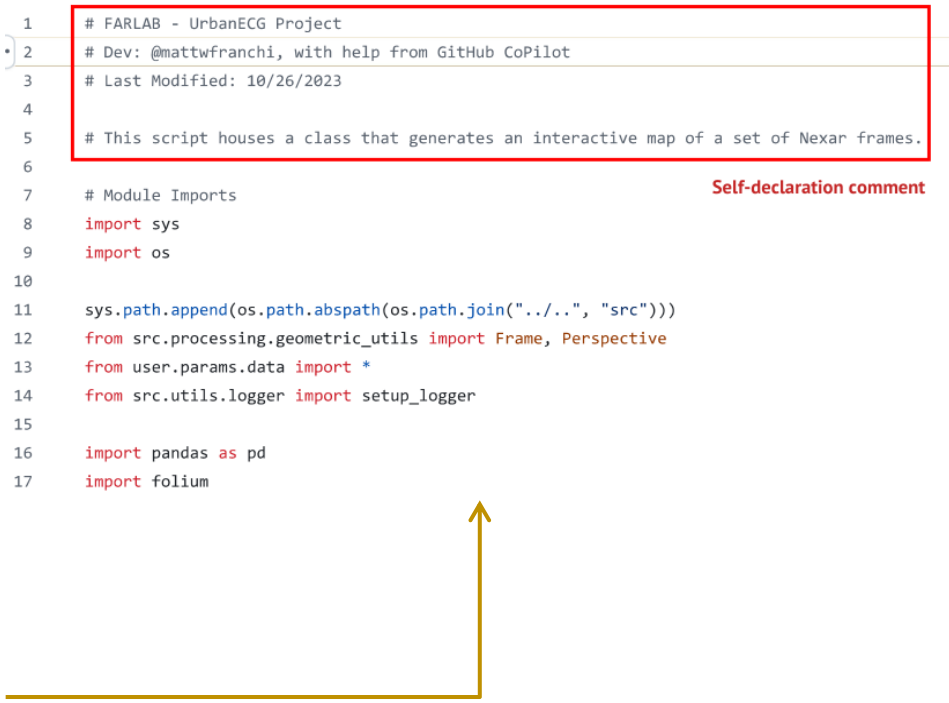
- Collaboration
- Speed
- Unpublished / negative results
- Re-use
- Learn by example
- Feedback

Outcomes of Openness

Let's talk about the use of AI

- Uploading data to AI tools (do not do).
- **Edinburgh Language Model (ELM):**
  - Explore the tool and the variety of models.
  - Guidance

- Follow university guidance: Using generative AI in your studies: guidelines for students
  - ALWAYS declare the use of AI.
  - If you have doubts, always check with your supervisors.
  - Examples:
    - *"I used GPT -5 via ELM to check grammar and spelling throughout my assignment."*
    - *"I also used Midjourney to generate the image on page 2."*
    - **For code:** Include model and version as a comment



Enduring Understanding

Planning ahead how you will manage your data – making data FAIR - is the cornerstone of data sharing

Openness can promote collaboration and career progression.

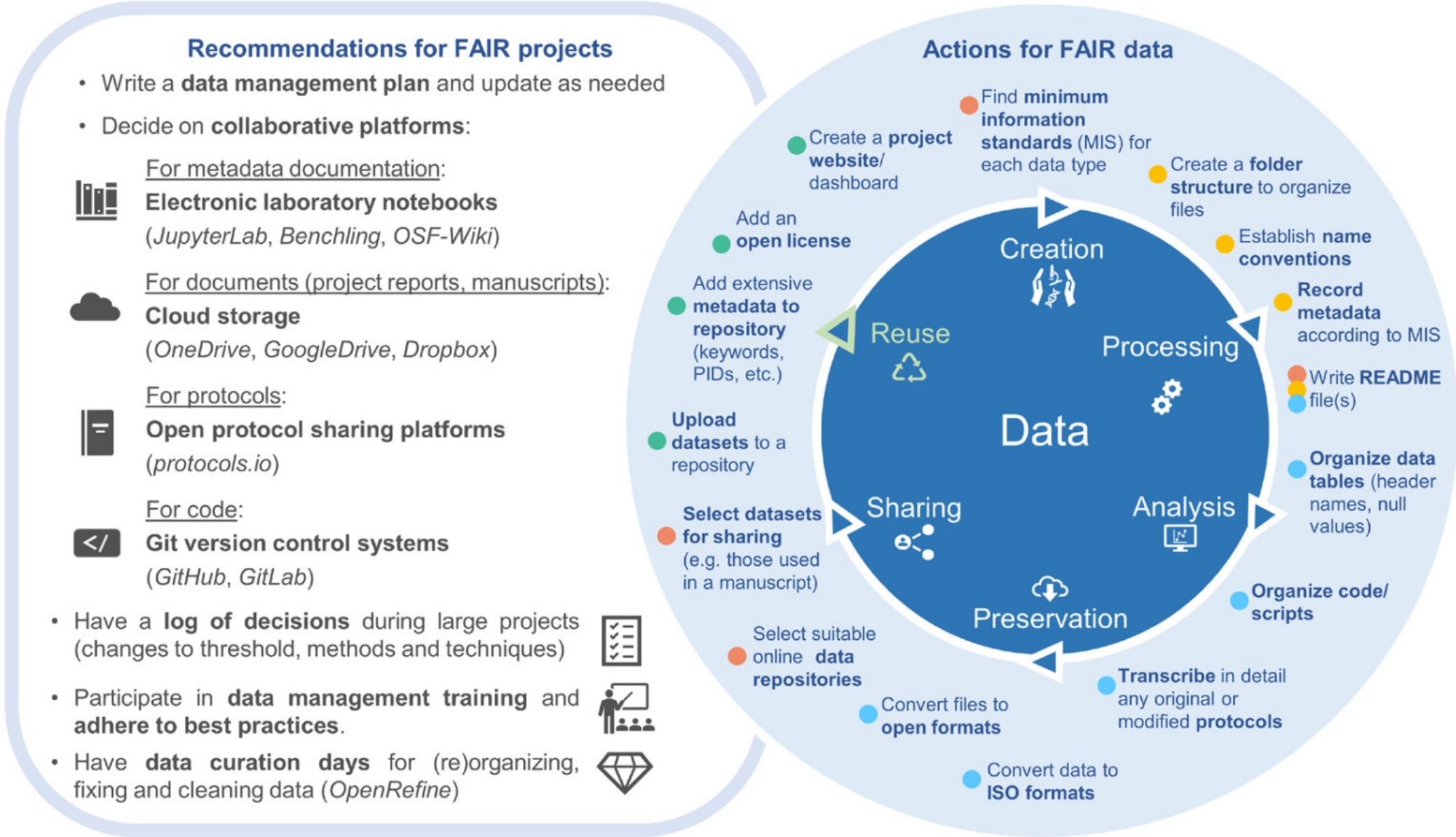


Practical Applications

- Make it easy to share data at the end of a project
- Support your CV by sharing meaningful outputs
- Contribute to data re-use and experimental reproducibility!

In Conclusion

Summary



Scorza LCT *et al.* Daily life in the Open Biologist's second job, as a Data Curator- Wellcome Open Res 2024, **9**:523 (https://doi.org/10.12688/wellcomeopenres.22899.1)



Data-related resources at the University

All the resources to manage your data through the PhD journey

- ✓ Storage
- ✓ Collaboration
- ✓ Code
- ✓ Protocols
- ✓ Repositories
- ✓ Additional Information

University resources for data management

BioRDM wiki
https://www.wiki.ed.ac.uk/display/RDMS

DRS page – Digital research services
https://digitalresearchservices.ed.ac.uk/

- Dr Burak Ozyurt – CSE Research Facilitator

RDS – Research data service page (UoE Central Service)
https://www.ed.ac.uk/information-services/research-support/research-data-service

University resources - Storage

UoE DataStore – for your active research data

- It looks like your computer disk (a network drive)
- Free (250GB + 250GB per student)
- Accessible from all computers
- Off-site backup and versioning/history of the files
- Easy sharing within the University
- Dropbox like access via DataSync

https://library.ed.ac.uk/research-support/research-data-service/during/data-storage

University resources - Storage

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Please Use DataStore

https://library.ed.ac.uk/research-support/research-data-service/during/data-storage

University resources - Storage

UoE DataStore – for your active research data

Two areas

- Group space (\\csce.datastore.ed.ac.uk\csce\biology\groups\YOUR-GROUP)
- Private (\\csce.datastore.ed.ac.uk\csce\biology\users\YOUR_LOGIN)

Use group space for data!

(request access through your PI and information services)

University resources - Storage

UoE DataStore – How to use

- You need VPN to connect from home

How to connect:

- https://www.ed.ac.uk/information-services/computing/desktop-personal/connect-uni-file-storage

University resources - Collaboration

Sharepoint / OneDrive (1TB)

https://uoe-my.sharepoint.com

- For documents, presentations, reports, etc.
- Excellent for collaborative work on shared documents
- Online versions of Word / Excel
- Your desktop office can directly sync with sharepoint files



- **DON’T bounce emails with your PI – co-edit on sharepoint!!!**

University resources - Collaboration

UoE WIKI

https://www.wiki.ed.ac.uk

- Projects' websites,
- Making notes, organizing files,
- Even a simple ELN (can be private or public)

https://www.wiki.ed.ac.uk/display/RDMS

University resources - Protocols

Protocols.io

https://www.protocols.io/

- private and public (creates DOI) – free version – 2 private protocols
- versioning / forking
- collaboration
- good for group SOPs

University resources - CODE

✓ If you code (script) you should use GIT
(you need to learn it though)

✓ Free GitHub (https://github.com)

✓ Free GitLab (supported by UoE - UoE GitLab)

University resources - Repositories

Choosing domain specific repository

- BioRDM wiki:

https://www.wiki.ed.ac.uk/spaces/RDMS/pages/424280992/Suggested+data+repositories

- Fairsharing.org

University resources - Repositories

UoE DataShare

https://datashare.ed.ac.uk/

- for sharing public datasets (possible embargo period)
- unlimited, free storage (250GB per dataset)
- DOI
- automatically creates records in your personal profile (UoE PURE)

Learning

Practical recipes and guides:

- https://faircookbook.elixir-europe.org/
- https://rdmkit.elixir-europe.org/

Training

- Ed-DASH https://edcarp.github.io/Ed-DaSH/index.html
- UoE RDS https://www.ed.ac.uk/information-services/research-support/research-data-service
- Digital Research Services: https://digitalresearchservices.ed.ac.uk/news-and-updates/



FEEDBACK

wooclap

Event code

ZPIIYMA

THE UNIVERSITY of EDINBURGH

Thank you!
Best of luck in your PhD journey

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